

Lecture 3.4: The Acceptance-Rejection Method

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Motivation

The inverse transformation method of Lecture 3.3 is elegant but has a real limitation: it needs the quantile function F_X^{-1} , and many important distributions — the Gamma distribution with non-integer shape, the standard Normal, most Bayesian posterior densities — have no closed-form quantile function at all. Numerically inverting the cdf at every single draw is possible but slow and can be delicate. The acceptance-rejection method sidesteps the problem entirely: it needs no quantile function whatsoever, only the ability to *evaluate* the target density (even up to an unknown constant) and to sample from some other, more convenient distribution. The idea originates with von Neumann's unpublished 1947 note on Monte Carlo methods for the Manhattan Project (von Neumann, 1951).

Intuition

Picture the graph of the target density f , sitting entirely beneath the graph of some scaled *proposal* density $Mg(x)$, where g is easy to sample from and $M \geq 1$ is large enough that Mg never dips below f . Suppose it is possible to scatter points uniformly at random in the two-dimensional region lying *under* the curve Mg . Among these scattered points, the ones that happen to also fall under the lower curve f are, geometrically, uniformly scattered under f — and a point picked uniformly from the region under a density curve has, as its horizontal coordinate, precisely a random draw from that density. Points falling above f (but still under Mg) are simply thrown away; the points that survive are exact draws from f , with no approximation involved.

Turning this picture into an algorithm requires a practical way to scatter points under Mg : draw the horizontal coordinate Y from g itself, then draw the vertical coordinate uniformly between 0 and $Mg(Y)$. This is equivalent to drawing an independent $U \sim \text{Uniform}(0, 1)$ and accepting the point exactly when $U \leq f(Y)/(Mg(Y))$ — a single numerical comparison, with no reference to areas or integrals needed while the algorithm is actually running.

Setup

Definition 4.1. Let f be a target density and g a *proposal* density from which sampling is easy, with a constant $M \geq 1$ such that

$$f(x) \leq Mg(x) \quad \text{for every } x.$$

Draw $Y \sim g$ and $U \sim \text{Uniform}(0, 1)$ independently; **accept** $X = Y$ if $U \leq f(Y)/(Mg(Y))$, otherwise discard both and repeat.

Correctness

Theorem 4.1. The accepted variate X has density exactly f , and each trial is accepted with probability $1/M$.

Proof. Conditional on $Y = y$, the acceptance probability is $f(y)/(Mg(y))$, a number in $[0, 1]$ because of the bound $f \leq Mg$ assumed in Definition 4.1. Averaging this acceptance probability over the distribution of Y gives the unconditional probability that any single trial succeeds:

$$P(\text{accept}) = \int \frac{f(y)}{Mg(y)} g(y) dy = \frac{1}{M} \int f(y) dy = \frac{1}{M},$$

using $\int f(y) dy = 1$ since f is a density. This proves the second claim. For the first claim, fix any x and compute the probability that a trial *both* has $Y \leq x$ *and* is accepted:

$$P(Y \leq x, \text{ accept}) = \int_{-\infty}^x \frac{f(y)}{Mg(y)} g(y) dy = \frac{F(x)}{M},$$

where F is the cdf corresponding to f . By the definition of conditional probability, the cdf of the *accepted* value X is

$$P(X \leq x) = P(Y \leq x \mid \text{accept}) = \frac{P(Y \leq x, \text{ accept})}{P(\text{accept})} = \frac{F(x)/M}{1/M} = F(x).$$

Since this holds for every x , X has cdf F , i.e. density f exactly — not approximately, and regardless of how loosely or tightly the envelope Mg was chosen (as long as $f \leq Mg$ holds everywhere). ■

Corollary. Because each trial is accepted independently with probability $1/M$, the number of trials needed to obtain one accepted value follows a Geometric($1/M$) distribution, with mean exactly M . This is the reason the constant M is worth minimizing: it is literally the average number of candidates that must be generated and discarded for every one that is kept.

Worked example. Let $f(x) = 6x(1-x)$ on $(0, 1)$ (a Beta(2, 2) density, with maximum value 1.5 at $x = 0.5$), and take g to be the Uniform(0, 1) density with $M = 1.5$. For a candidate $y = 0.9$, $f(0.9) = 6(0.9)(0.1) = 0.54$ and $Mg(0.9) = 1.5$, so the acceptance ratio is $0.54/1.5 = 0.36$. A draw $u = 0.5$ exceeds 0.36, so this candidate is **rejected**; a draw $u = 0.2$ does not exceed 0.36, so the same candidate would be **accepted**. Note that $y = 0.9$ lies far from the density's peak at 0.5, which is exactly why its acceptance ratio of 0.36 is well below 1 — candidates near the peak of f are accepted far more readily than candidates in the tails.

Choice of M and Exponential Tilting

By the corollary above, the average computational cost per accepted value is exactly M , so the natural design goal is to make M as small as possible. The smallest valid constant is $M^* = \sup_x f(x)/g(x)$; achieving a finite M^* at all requires g 's support and tails to be at least as generous as f 's — for instance, a Normal proposal can never bound a Cauchy target, because the Cauchy density decays far more slowly than the Normal density does in the tails, so the ratio $f(x)/g(x)$ grows without bound.

Exponential tilting is a systematic way to build a good proposal when the region of interest is a distribution's tail. It chooses a proposal from the family $g_\theta(x) \propto e^{\theta x} h(x)$, for some convenient base density h , and tunes the “tilt” θ so that g_θ concentrates its mass exactly where the target region lies. As a concrete instance, to sample the standard Normal tail $\{Z > a\}$ efficiently for large a , take h to be the Exponential density shifted to start at a , with rate $\theta = a$; a short calculation (completing the square in the exponent of the Normal density) shows the resulting bounding constant is

$$M^* = \frac{\varphi(a)}{a Q(a)}, \quad Q(a) = P(Z > a),$$

and by the classical Mills' ratio approximation $Q(a) \sim \varphi(a)/a$ for large a , this constant satisfies $M^* \rightarrow 1$ as $a \rightarrow \infty$. In words: the tilted method becomes *more* efficient the further out into the tail one simulates, which is the exact opposite of naive sampling from the untilted Normal distribution and discarding values below

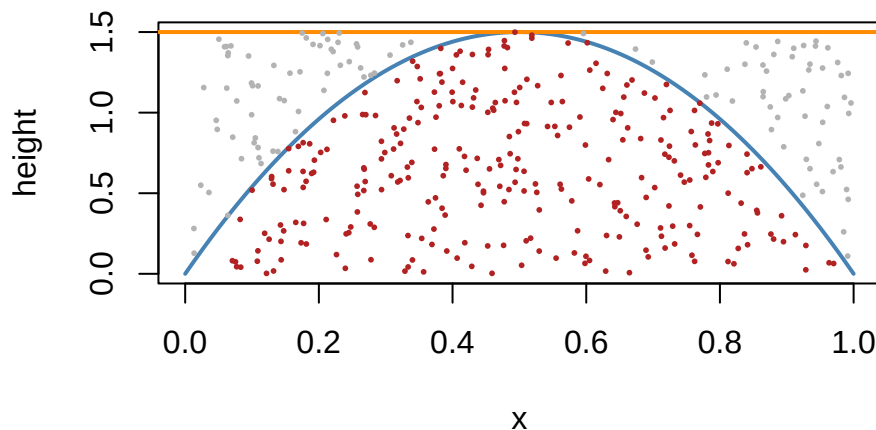


Figure 1: Acceptance-rejection for $f(x) = 6x(1-x)$ using a $\text{Uniform}(0,1)$ envelope, $M = 1.5$; accepted points (red) fall under f , rejected points (grey) lie above it.

the threshold — an approach whose acceptance probability $Q(a)$ shrinks toward zero as a grows, making it hopelessly wasteful for exactly the extreme, rare events that are often of the greatest practical interest.

Numerical illustration. At threshold $a = 2$, $\varphi(2) = 0.05399$ and $Q(2) = P(Z > 2) = 0.02275$, giving $M^* = 0.05399/(2 \times 0.02275) = 1.187$, so the tilted method accepts a candidate with probability $1/M^* \approx 0.843$. The naive method — drawing an ordinary standard Normal value and keeping it only if it exceeds 2 — accepts with probability $Q(2) = 0.02275$ directly. The tilted method is therefore roughly $0.843/0.02275 \approx 37$ times more efficient at this threshold alone, and the gap widens further as a increases.

References

von Neumann, J. (1951). Various techniques used in connection with random digits. In *Monte Carlo Method* (National Bureau of Standards Applied Mathematics Series 12, pp. 36–38). U.S. Government Printing Office.